DBMS-7-10

https://github.com/mrudulabbca25-crypto/DBMS--7-10.git

```
AFTER UPDATE ON Employee
FOR EACH ROW
BEGIN
    INSERT INTO Employee_Log(message)
    VALUES ('Employee updated');
END //

DELIMITER ;

DELIMITER //

CREATE TRIGGER before_employee_delete
BEFORE DELETE ON Employee
FOR EACH ROW
BEGIN
    IF OLD.emp_id = 1 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Employee cannot be deleted';
    END IF;
END //

DELIMITER ;

DELIMITER //

CREATE TRIGGER after_employee_delete
AFTER DELETE ON Employee
FOR EACH ROW
BEGIN
    INSERT INTO Employee_Log(message)
    VALUES ('Employee deleted');
END //

DELIMITER ;

INSERT INTO Employee VALUES
(7, 'Rohan', 'HR', -5000);

UPDATE Employee
SET salary = -1000
```

```
WHERE emp_id = 2;

UPDATE Employee
SET salary = 60000
WHERE emp_id = 3;

DELETE FROM Employee
WHERE emp_id = 7;

SELECT * FROM Employee;

SELECT * FROM Employee_Log;
```

OUPUT:

```
+--------+-------+------------+--------+
| emp_id | name  | department | salary |
+--------+-------+------------+--------+
|      1 | Rahul | IT         |  50000 |
|      2 | Priya | HR         |  40000 |
|      3 | Aman  | IT         |  60000 |
|      4 | Neha  | HR         |  45000 |
|      5 | Karan | Sales      |  35000 |
+--------+-------+------------+--------+
+--------+-------+------------+--------+
| emp_id | name  | department | salary |
+--------+-------+------------+--------+
|      1 | Rahul | IT         |  55000 |
|      2 | Priya | HR         |      0 |
|      3 | Aman  | IT         |  60000 |
|      4 | Neha  | HR         |  45000 |
|      5 | Karan | Sales      |  35000 |
+--------+-------+------------+--------+
+--------+------------------+---------------------+
| log_id | message          | log_time            |
+--------+------------------+---------------------+
|      1 | Employee inserted | 2026-10-04 08:51:47 |
|      2 | Employee updated  | 2026-10-04 08:51:47 |
|      3 | Employee updated  | 2026-10-04 08:51:47 |
|      4 | Employee deleted  | 2026-10-04 08:51:47 |
+--------+------------------+---------------------+
```

```
mysql>
mysql> SELECT * FROM Student;
+------------+--------------+---------+-------+
| Student_ID | Student_Name | Dept_ID | Marks |
+------------+--------------+---------+-------+
|        201 | Aditya       |      11 |    82 |
|        202 | Kavya        |      22 |    91 |
|        203 | Rohit        |      11 |    78 |
|        204 | Ishita       |      33 |    87 |
|        205 | Manoj        |      22 |    94 |
|        206 | Divya        |      55 |    73 |
+------------+--------------+---------+-------+
6 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM Department;
+---------+---------------+-----------+
| Dept_ID | Department    | HOD       |
+---------+---------------+-----------+
|      11 | AI & DS       | Dr. Kumar |
|      22 | Mechanical    | Dr. Patel |
|      33 | EEE           | Dr. Iyer  |
|      44 | Biotechnology | Dr. Verma |
+---------+---------------+-----------+
4 rows in set (0.00 sec)

mysql>
mysql>
mysql> -- 1. INNER JOIN
mysql> -- Returns only matching records from both tables.
mysql>
mysql> SELECT
    ->     Student.Student_ID,
    ->     Student.Student_Name,
    ->     Student.Dept_ID,
    ->     Department.Department,
    ->     Department.HOD,
    ->     Student.Marks
    -> FROM Student
    -> INNER JOIN Department
    -> ON Student.Dept_ID = Department.Dept_ID;
```

```
mysql>
mysql>
mysql> -- 3. RIGHT JOIN
mysql> -- Returns all records from the right table and matching records from the left table.
mysql>
mysql> SELECT
    ->     Student.Student_ID,
    ->     Student.Student_Name,
    ->     Student.Dept_ID,
    ->     Department.Department,
    ->     Department.HOD,
    ->     Student.Marks
    -> FROM Student
    -> RIGHT JOIN Department
    -> ON Student.Dept_ID = Department.Dept_ID;
+------------+--------------+---------+---------------+-----------+-------+
| Student_ID | Student_Name | Dept_ID | Department    | HOD       | Marks |
+------------+--------------+---------+---------------+-----------+-------+
|        203 | Rohit        |      11 | AI & DS       | Dr. Kumar |    78 |
|        201 | Aditya       |      11 | AI & DS       | Dr. Kumar |    82 |
|        205 | Manoj        |      22 | Mechanical    | Dr. Patel |    94 |
|        202 | Kavya        |      22 | Mechanical    | Dr. Patel |    91 |
|        204 | Ishita       |      33 | EEE           | Dr. Iyer  |    87 |
|       NULL | NULL         |    NULL | Biotechnology | Dr. Verma |  NULL |
+------------+--------------+---------+---------------+-----------+-------+
6 rows in set (0.00 sec)
```

```
+-------+-------+
| Name  | Marks |
+-------+-------+
| Aarav |    84 |
+-------+-------+
1 row in set (0.00 sec)

mysql>
mysql>
mysql> SELECT Name, Department
    -> FROM Student
    -> WHERE Department IN (
    ->     SELECT Department
    ->     FROM Student
    ->     WHERE City = 'Mysore'
    -> );
+---------+------------+
| Name    | Department |
+---------+------------+
| Diya    | Mechanical |
| Karthik | Mechanical |
+---------+------------+
2 rows in set (0.00 sec)

mysql>
mysql>
mysql> SELECT Name, Marks
    -> FROM Student
    -> WHERE Marks > ALL (
    ->     SELECT Marks
    ->     FROM Student
    ->     WHERE Department = 'EEE'
    -> );
+---------+-------+
| Name    | Marks |
+---------+-------+
| Diya    |    91 |
| Karthik |    95 |
+---------+-------+
2 rows in set (0.00 sec)
```

```
+---------+-------+
| Name    | Marks |
+---------+-------+
| Diya    |    91 |
| Ananya  |    88 |
| Karthik |    95 |
| Varun   |    86 |
+---------+-------+
4 rows in set (0.00 sec)

mysql>
mysql>
mysql> SELECT Name, Department
    -> FROM Student
    -> WHERE Department = (
    ->     SELECT Department
    ->     FROM Student
    ->     WHERE Name = 'Aarav'
    -> );
+---------+------------+
| Name    | Department |
+---------+------------+
| Aarav   | AI & DS    |
| Rishabh | AI & DS    |
| Ishita  | AI & DS    |
+---------+------------+
3 rows in set (0.00 sec)

mysql>
mysql>
mysql> SELECT Name, Marks
    -> FROM Student
    -> WHERE Department = 'AI & DS'
    -> AND Marks > (
    ->     SELECT AVG(Marks)
    ->     FROM Student
    ->     WHERE Department = 'AI & DS'
    -> );
```

```
1 row in set (0.00 sec)

mysql>
mysql>
mysql> SELECT Name, Marks
    -> FROM Student
    -> WHERE Marks > (
    ->     SELECT AVG(Marks)
    ->     FROM Student
    -> );
+---------+-------+
| Name    | Marks |
+---------+-------+
| Diya    |    91 |
| Ananya  |    88 |
| Karthik |    95 |
| Varun   |    86 |
+---------+-------+
4 rows in set (0.00 sec)

mysql>
mysql>
mysql> SELECT Name, Marks
    -> FROM Student
    -> WHERE Marks = (
    ->     SELECT MAX(Marks)
    ->     FROM Student
    -> );
+---------+-------+
| Name    | Marks |
+---------+-------+
| Karthik |    95 |
+---------+-------+
1 row in set (0.00 sec)

mysql>
mysql>
mysql> SELECT Name, Marks
    -> FROM Student
    -> WHERE Marks > (
    ->     SELECT Marks
    ->     FROM Student
    ->     WHERE Name = 'Aarav'
    -> );
```

```
+---------+-------+
| Name    | Marks |
+---------+-------+
| Aarav   |    84 |
| Rishabh |    77 |
| Ishita  |    71 |
+---------+-------+
3 rows in set (0.00 sec)

mysql>
mysql>
mysql> SELECT Name, Marks
    -> FROM Student
    -> WHERE City = 'Coimbatore'
    -> AND Marks > 80;
+-------+-------+
| Name  | Marks |
+-------+-------+
| Aarav |    84 |
+-------+-------+
1 row in set (0.00 sec)

mysql>
mysql>
mysql> SELECT MAX(Marks) AS Highest_Marks
    -> FROM Student;
+---------------+
| Highest_Marks |
+---------------+
|            95 |
+---------------+
1 row in set (0.00 sec)

mysql>
mysql>
mysql> SELECT AVG(Marks) AS Average_Marks
    -> FROM Student;
+---------------+
| Average_Marks |
+---------------+
|       84.5714 |
+---------------+
```

```
mysql>
mysql>
mysql> SELECT Name, Department, Marks
    -> FROM Student;
+---------+------------+-------+
| Name    | Department | Marks |
+---------+------------+-------+
| Aarav   | AI & DS    |    84 |
| Diya    | Mechanical |    91 |
| Rishabh | AI & DS    |    77 |
| Ananya  | EEE        |    88 |
| Karthik | Mechanical |    95 |
| Ishita  | AI & DS    |    71 |
| Varun   | EEE        |    86 |
+---------+------------+-------+
7 rows in set (0.00 sec)

mysql>
mysql>
mysql> SELECT Name, Marks
    -> FROM Student
    -> WHERE Marks > 80;
+---------+-------+
| Name    | Marks |
+---------+-------+
| Aarav   |    84 |
| Diya    |    91 |
| Ananya  |    88 |
| Karthik |    95 |
| Varun   |    86 |
+---------+-------+
5 rows in set (0.00 sec)

mysql>
mysql>
mysql> SELECT Name, Marks
    -> FROM Student
    -> WHERE Department = 'AI & DS';
```

```
mysql> CREATE DATABASE CollegeStudentDB;
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> USE CollegeStudentDB;
Database changed
mysql>
mysql> CREATE TABLE Student (
    ->     Student_ID INT PRIMARY KEY,
    ->     Name VARCHAR(50),
    ->     Department VARCHAR(20),
    ->     Marks INT,
    ->     City VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> INSERT INTO Student VALUES
    -> (301, 'Aarav', 'AI & DS', 84, 'Coimbatore'),
    -> (302, 'Diya', 'Mechanical', 91, 'Mysore'),
    -> (303, 'Rishabh', 'AI & DS', 77, 'Kochi'),
    -> (304, 'Ananya', 'EEE', 88, 'Madurai'),
    -> (305, 'Karthik', 'Mechanical', 95, 'Pune'),
    -> (306, 'Ishita', 'AI & DS', 71, 'Salem'),
    -> (307, 'Varun', 'EEE', 86, 'Trichy');
Query OK, 7 rows affected (0.01 sec)
Records: 7  Duplicates: 0  Warnings: 0

mysql>
mysql>
mysql> SELECT *
    -> FROM Student;
+------------+---------+------------+-------+------------+
| Student_ID | Name    | Department | Marks | City       |
+------------+---------+------------+-------+------------+
|        301 | Aarav   | AI & DS    |    84 | Coimbatore |
|        302 | Diya    | Mechanical |    91 | Mysore     |
|        303 | Rishabh | AI & DS    |    77 | Kochi      |
|        304 | Ananya  | EEE        |    88 | Madurai    |
|        305 | Karthik | Mechanical |    95 | Pune       |
|        306 | Ishita  | AI & DS    |    71 | Salem      |
|        307 | Varun   | EEE        |    86 | Trichy     |
+------------+---------+------------+-------+------------+
```

```
mysql>
mysql>
mysql> -- 4. FULL OUTER JOIN
mysql> -- Returns all records from both tables.
mysql>
mysql> SELECT
    ->     Student.Student_ID,
    ->     Student.Student_Name,
    ->     Student.Dept_ID,
    ->     Department.Department,
    ->     Department.HOD,
    ->     Student.Marks
    -> FROM Student
    -> LEFT JOIN Department
    -> ON Student.Dept_ID = Department.Dept_ID
    ->
    -> UNION
    ->
    -> SELECT
    ->     Student.Student_ID,
    ->     Student.Student_Name,
    ->     Student.Dept_ID,
    ->     Department.Department,
    ->     Department.HOD,
    ->     Student.Marks
    -> FROM Student
    -> RIGHT JOIN Department
    -> ON Student.Dept_ID = Department.Dept_ID;
+------------+--------------+---------+---------------+-----------+-------+
| Student_ID | Student_Name | Dept_ID | Department    | HOD       | Marks |
+------------+--------------+---------+---------------+-----------+-------+
|        201 | Aditya       |      11 | AI & DS       | Dr. Kumar |    82 |
|        202 | Kavya        |      22 | Mechanical    | Dr. Patel |    91 |
|        203 | Rohit        |      11 | AI & DS       | Dr. Kumar |    78 |
|        204 | Ishita       |      33 | EEE           | Dr. Iyer  |    87 |
|        205 | Manoj        |      22 | Mechanical    | Dr. Patel |    94 |
|        206 | Divya        |      55 | NULL          | NULL      |    73 |
|       NULL | NULL         |    NULL | Biotechnology | Dr. Verma |  NULL |
+------------+--------------+---------+---------------+-----------+-------+
7 rows in set (0.01 sec)
```

```
mysql>
mysql>
mysql> -- 3. RIGHT JOIN
mysql> -- Returns all records from the right table and matching records from the left table.
mysql>
mysql> SELECT
    ->     Student.Student_ID,
    ->     Student.Student_Name,
    ->     Student.Dept_ID,
    ->     Department.Department,
    ->     Department.HOD,
    ->     Student.Marks
    -> FROM Student
    -> RIGHT JOIN Department
    -> ON Student.Dept_ID = Department.Dept_ID;
+------------+--------------+---------+---------------+-----------+-------+
| Student_ID | Student_Name | Dept_ID | Department    | HOD       | Marks |
+------------+--------------+---------+---------------+-----------+-------+
|        203 | Rohit        |      11 | AI & DS       | Dr. Kumar |    78 |
|        201 | Aditya       |      11 | AI & DS       | Dr. Kumar |    82 |
|        205 | Manoj        |      22 | Mechanical    | Dr. Patel |    94 |
|        202 | Kavya        |      22 | Mechanical    | Dr. Patel |    91 |
|        204 | Ishita       |      33 | EEE           | Dr. Iyer  |    87 |
|       NULL | NULL         |    NULL | Biotechnology | Dr. Verma |  NULL |
+------------+--------------+---------+---------------+-----------+-------+
6 rows in set (0.00 sec)
```

```
mysql>
mysql>
mysql> -- 2. LEFT JOIN
mysql> -- Returns all records from the left table and matching records from the right table.
mysql>
mysql> SELECT
    ->     Student.Student_ID,
    ->     Student.Student_Name,
    ->     Student.Dept_ID,
    ->     Department.Department,
    ->     Department.HOD,
    ->     Student.Marks
    -> FROM Student
    -> LEFT JOIN Department
    -> ON Student.Dept_ID = Department.Dept_ID;
+------------+--------------+---------+------------+-----------+-------+
| Student_ID | Student_Name | Dept_ID | Department | HOD       | Marks |
+------------+--------------+---------+------------+-----------+-------+
|        201 | Aditya       |      11 | AI & DS    | Dr. Kumar |    82 |
|        202 | Kavya        |      22 | Mechanical | Dr. Patel |    91 |
|        203 | Rohit        |      11 | AI & DS    | Dr. Kumar |    78 |
|        204 | Ishita       |      33 | EEE        | Dr. Iyer  |    87 |
|        205 | Manoj        |      22 | Mechanical | Dr. Patel |    94 |
|        206 | Divya        |      55 | NULL       | NULL      |    73 |
+------------+--------------+---------+------------+-----------+-------+
6 rows in set (0.00 sec)
```